

1. Task Overview

Task Title:

Setting up a Virtual Lab + File Integrity Monitoring using Wazuh

Task Context:

This report documents the setup of a Wazuh SIEM environment and the subsequent triage of security alerts generated during testing, specifically focusing on file integrity and authentication failures.

2. Objective

To configure the Wazuh Manager and Agent, verify telemetry flow, and demonstrate the detection of unauthorized file changes and brute-force attempts.

3. Alert Summary

Field	Sample Value
Alert Name	sshd: Attempt to login using a non-existent user
Alert Time	Apr 20, 2026 @ 18:51:48.441
Affected User	admin
Host	kali(Agent 001)
Source IP	10.0.2.15
Initial Severity	Medium

4. Observed Indicators

- Rule ID 5710: Multiple attempts to log in as an admin with the wrong password were recorded.
- Rule ID 554: A new file (/etc/wazuh_test_again.txt) was detected in a restricted system directory.

5. Triage Questions and Answers

Question	Analyst Note
What happened?	A series of unauthorized file creations were followed by sequence of failed login attempts
Where did it happen?	SSH attack activity was recorded on wazuh-server. File integrity event was recorded on kali agent (Agent 001).
Who was involved?	User admin on Wazuh - server from source ip 10.0.2.15 Kali Linux
Is the behavior expected?	No. These actions simulate unauthorized attempts.
What is missing?	Network traffic detection

6. Evidence Section

Figure 1. Alert view in dashboard

W.

Threat Hunting

a

162 hits

Apr 19, 2026 @ 18:45:09.790 - Apr 20, 2026 @ 18:45:09.790

Export Formatted

Reset view

633 available fields

Columns

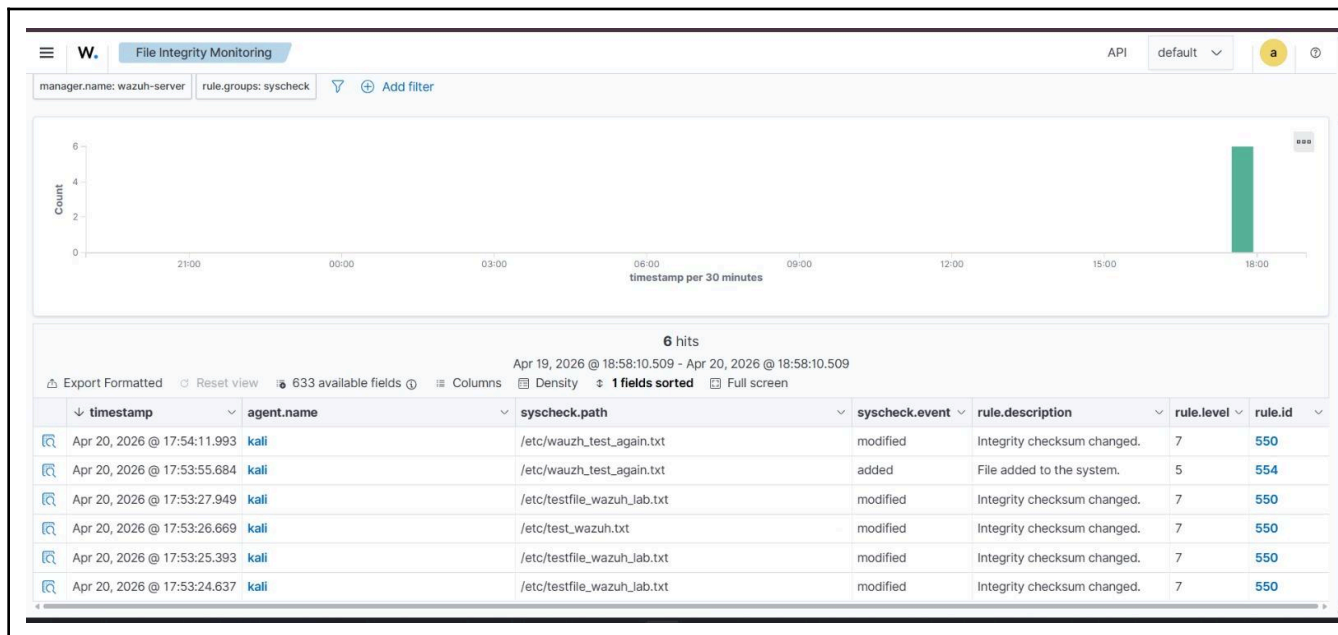
Density

1 fields sorted

Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
 Apr 20, 2026 @ 18:41:23.790	wazuh-server	sshd: Attempt to login using a non-existent user	5	5710
 Apr 20, 2026 @ 18:41:17.787	wazuh-server	sshd: Attempt to login using a non-existent user	5	5710
 Apr 20, 2026 @ 18:41:15.785	wazuh-server	PAM: User login failed.	5	5503
 Apr 20, 2026 @ 18:41:11.783	wazuh-server	sshd: Attempt to login using a non-existent user	5	5710
 Apr 20, 2026 @ 18:39:19.683	wazuh-server	sshd: Attempt to login using a non-existent user	5	5710
 Apr 20, 2026 @ 18:39:15.680	wazuh-server	sshd: Attempt to login using a non-existent user	5	5710
 Apr 20, 2026 @ 18:39:11.676	wazuh-server	PAM: User login failed.	5	5503
 Apr 20, 2026 @ 18:39:09.675	wazuh-server	sshd: Attempt to login using a non-existent user	5	5710
 Apr 20, 2026 @ 18:38:40.743	kali	PAM: Login session closed.	3	5502
 Apr 20, 2026 @ 18:38:00.769	kali	PAM: Login session opened.	3	5501
 Apr 20, 2026 @ 18:38:00.711	kali	Successful sudo to ROOT executed.	3	5402
 Apr 20, 2026 @ 18:31:25.126	wazuh-server	PAM: Login session closed.	3	5502
 Apr 20, 2026 @ 18:30:56.352	kali	PAM: Login session closed.	3	5502
 Apr 20, 2026 @ 18:30:20.410	kali	PAM: Login session opened.	3	5501
 Apr 20, 2026 @ 18:30:20.359	kali	Successful sudo to ROOT executed.	3	5402

Figure 2. File Integrity Monitoring



7. Analyst Decision

The alerts should be treated as suspicious. During testing, I noticed that the system flagged a file being added to a protected folder (/etc) and multiple failed attempts to log in with a fake username. Because these things don't usually happen during normal computer use, they indicate that someone is trying to test the system's defenses. It is not a "False Positive" because the SIEM correctly identified the unauthorized actions I performed.

8. Recommended Next Action

1. Validate with the user whether the login activity was legitimate.
2. Check whether the source IP has appeared in previous authentication logs.
3. Review the account for additional suspicious actions after the successful login.
4. Harden SSH: Disable root login and implement public-key authentication.

9. Short Analyst Summary

In this task, I successfully connected a Kali Linux machine to the Wazuh SIEM. I verified that the system is working by triggering two types of alerts: File Integrity (detecting a new file) and Authentication Failure. While the system is great at catching things that leave a "log" (like a password attempt), I learned that it doesn't automatically see "silent" network activity like an Nmap scan. Overall, the SIEM is properly configured to monitor the host's safety.